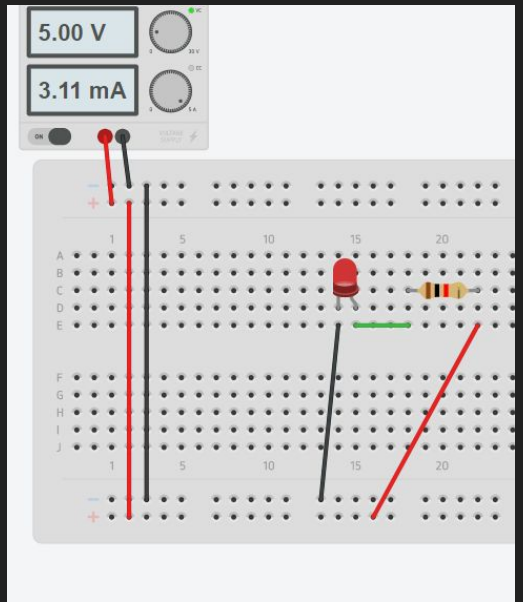
Digital Electronics Interfacing

Lesson 2

Current

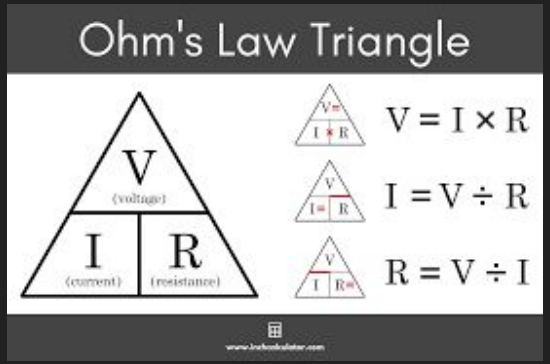
Current is measure in __amperes_____.
__Ampere_____ for short, is defined as the flow of one coulomb of charge per second, or **6.2 × 10^18** electrons per second.



Question:
What is the current in this circuit?
0.00311A
How many electrons are flowing through the LED every second?

1.9282×10^{16}

Ohm’s Law



We use Ohm’s Law to calculate the Voltage, Resistance and Current in a circuit. We only need the value of two of these to calculate the third. Voltage is measured in __volts_____, resistance in ____ohms_____ and current in ___amps_____.

Question:

To calculate current in a circuit what do you need to know?

I=V/R

Calculating the Current

The voltage in the circuit is 5V. The resistance is 1000Ω. What is the current?

According to Ohms Law

__V = I × R_____

There are __1000_____ milliAmps in 1 Amp so we multiplied 0.005A by __1000_____ to convert it to milliAmps. We would have done the same thing to go from Kilograms to milligrams.



More Accurate Results?

Most LEDs have an internal resistance of 13mΩ so what is the actual current in a circuit with just an LED and no other resistor?

____384.6A_____

That is an extremely large current. Most LEDs can only tolerate 10-30mA!!! That's why we need a resistor to reduce the current.

Note that you should always convert to base units i.e. mΩ to Ω or KV to V or mA to A etc.

Optimal LED Resistor Value?

Most LEDs are designed to operate optimally at about 10-30mA. Let's say we want a circuit to have 15mA of current. What should the resistor's value be in the circuit?.

_____330 Ω

_____

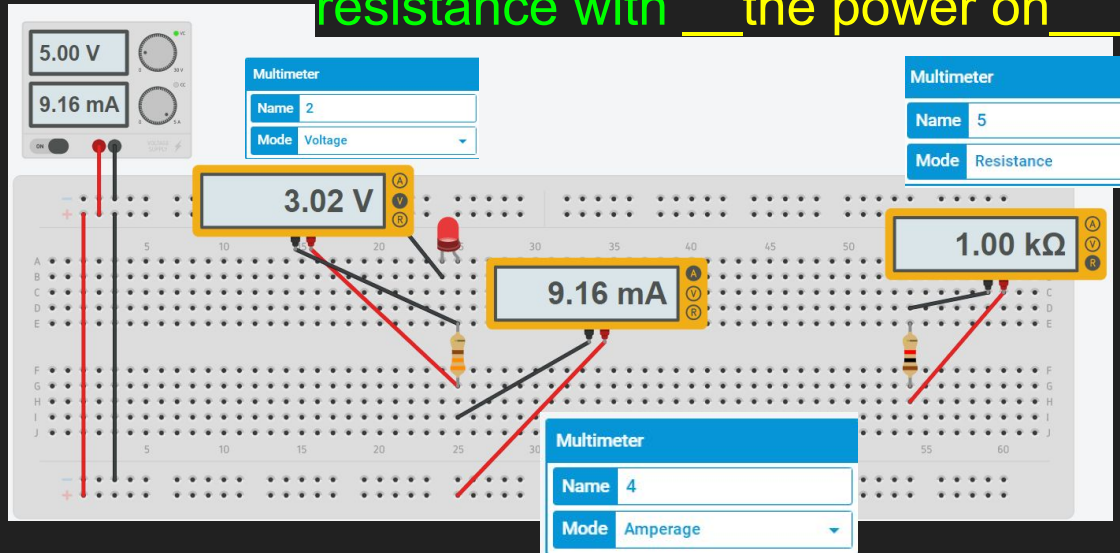
This demonstrates that inserting a resistor with a 300-400Ω value should be standard practice when using an LED in a 5V circuit.

Question:
What should the resistance be in the circuit if the voltage was doubled?
666.7Ω

Measuring Resistance, Current and Voltage in a Circuit



To measure the resistance in a circuit we can use a multimeter. To do this in Tinkercad you add a multimeter to the circuit. Change the option to Ω and then connect the two wires from the multimeter across any two points whose resistance you want to find (never measure resistance with __the power on_____



Question:

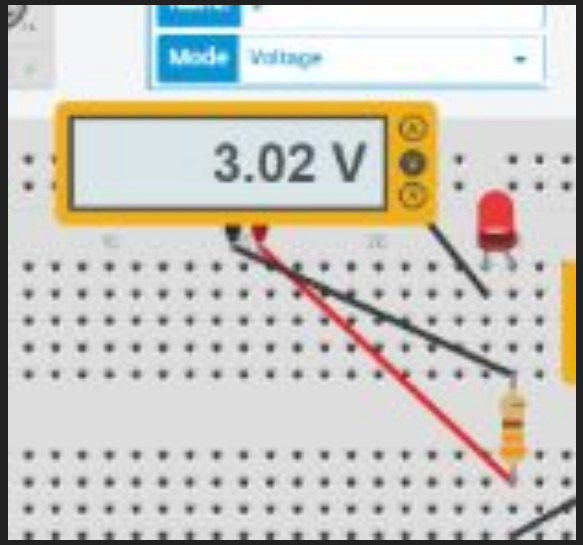
What is the resistance in the resistor?

__220Ω_____

What else is the multimeter being used to measure in this circuit?

__current_____

Measuring Voltage

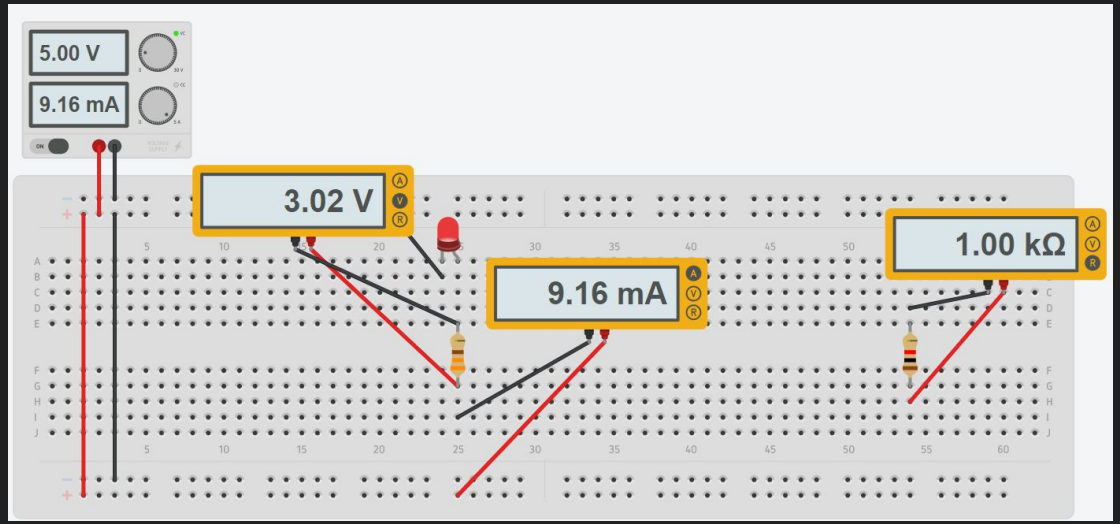


When you use a multimeter to measure the voltage you place the wires across any two points in the circuit you want to measure the voltage drop for. Every time electrons pass through something with resistance the voltaic pressure in the circuit drops and that's what you are measuring.

In the example circuit the total voltage is 5V but the voltage drops ___3.02_____ V across the resistor and will drop an additional ___1.98_____ V across the LED to give us a total drop across the whole circuit of 5V. You can see this by placing the multimeter wires at the source of power and the other wire at ground (GND).

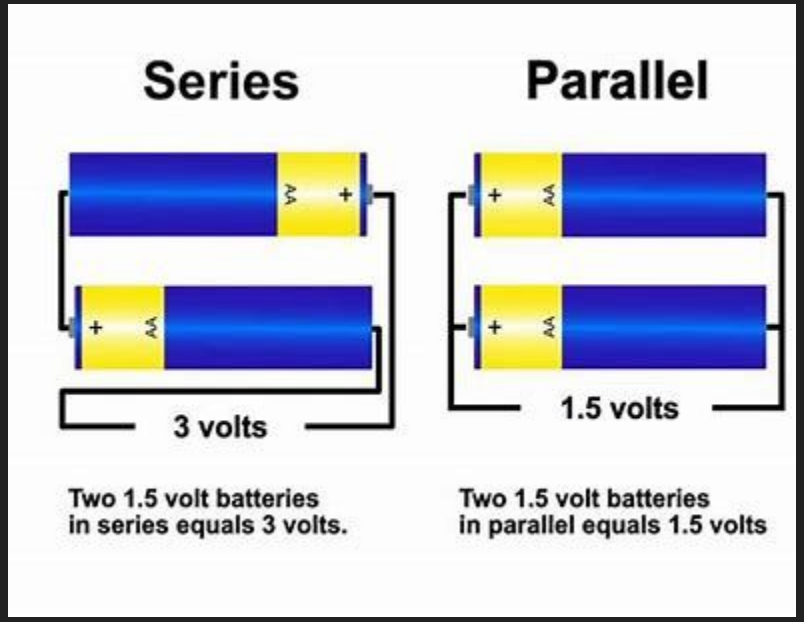
Measuring Current

When you use a multimeter to measure the current in a circuit you must insert the wires of the multimeter to become part of the path of electrons. Do not place the wires across any component like you would for measuring voltage drops or resistance.



Notice in this circuit the electrons must flow through the multimeter to complete the circuit. Notice also that we have not placed the multimeter wires across any component.

Voltage in Series



When batteries are connected in series i.e. electrons must flow through one to get through the other, the total voltage in the circuit is the sum of the voltages in all batteries.

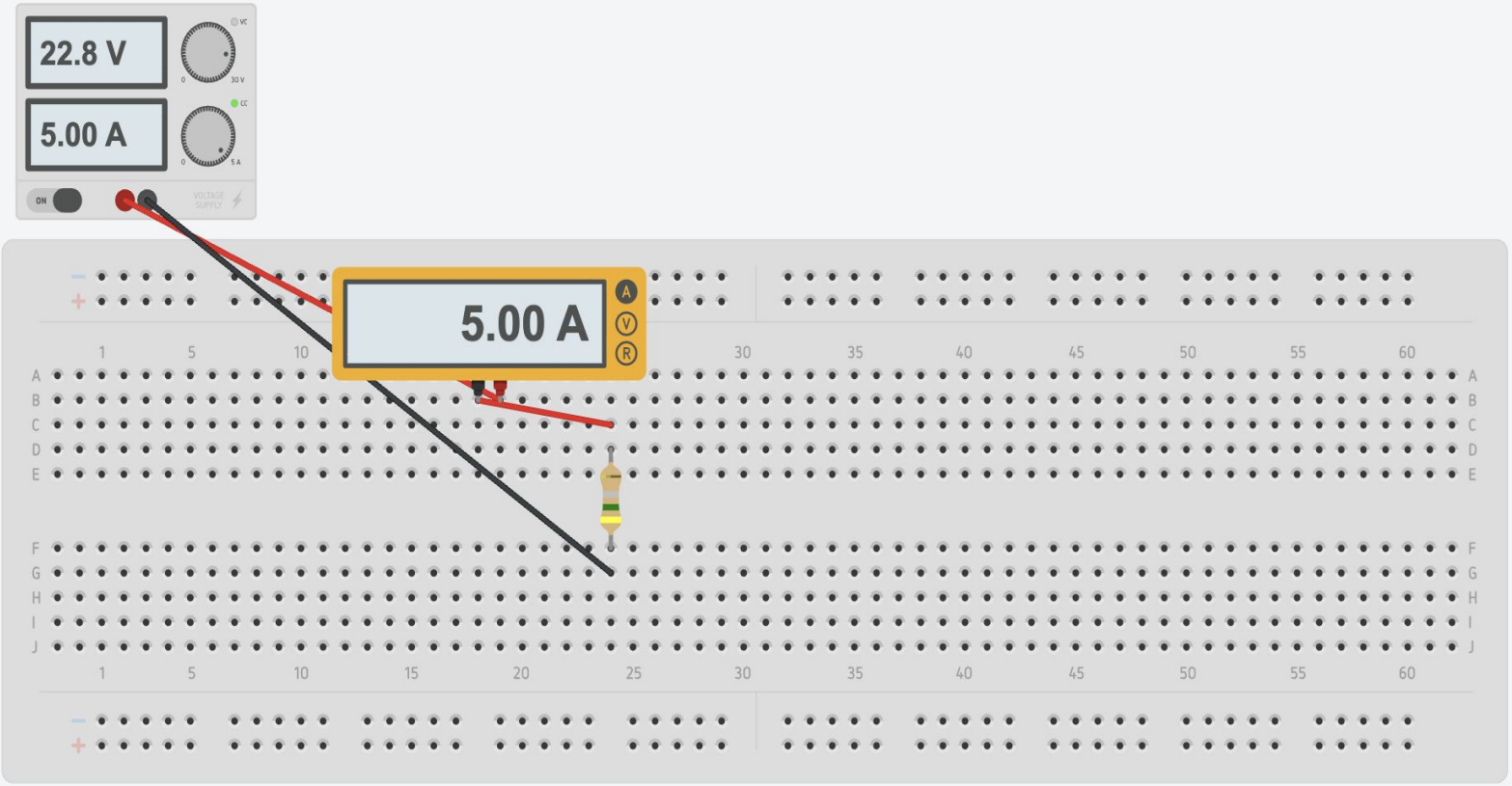
In parallel, where electrons have multiple paths back to GND, the voltage remains the same.

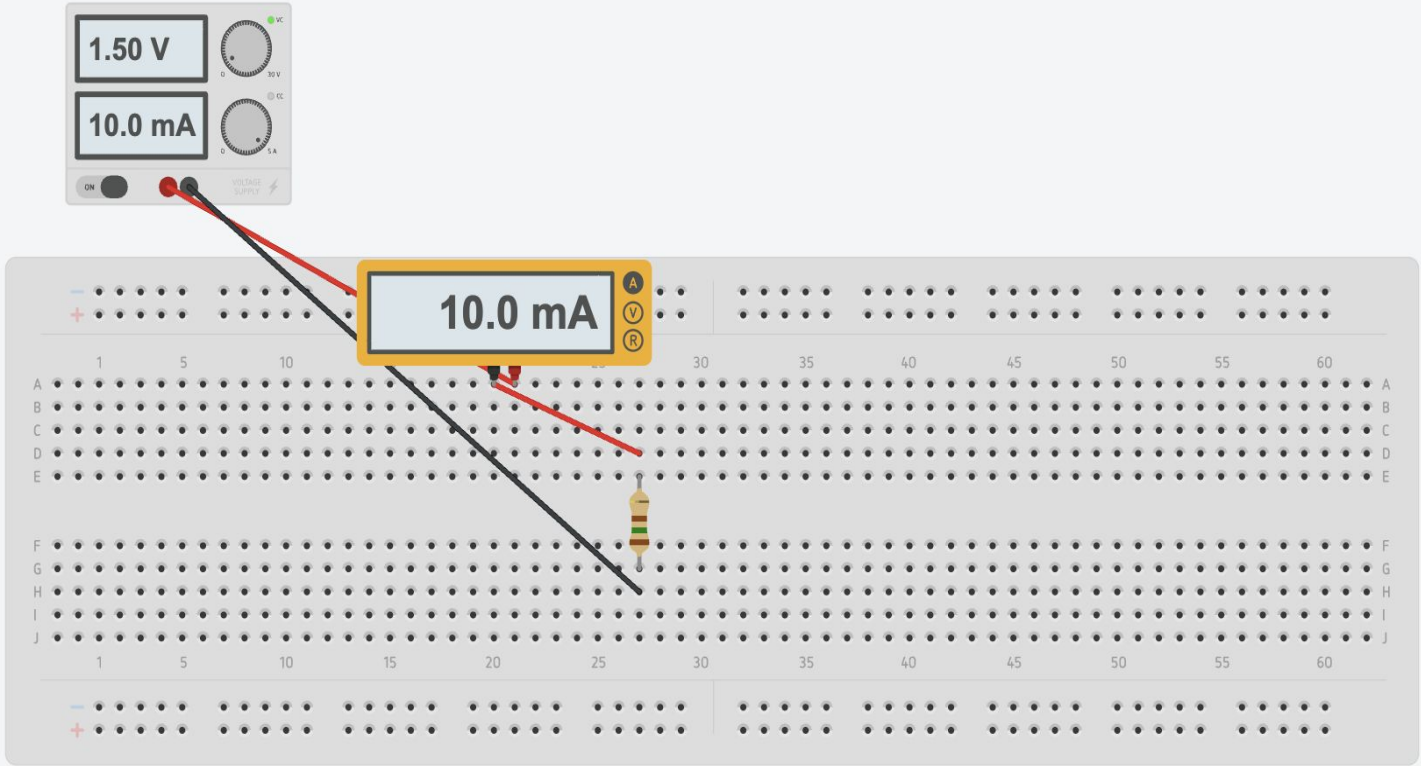
Questions and Exercises

1. Complete the this worksheet:
   https://docs.google.com/document/d/14wSEI4PICwP1VQ6UEJhftawc4mGvI1dws21VA-zIDxg/edit?usp=sharing

   https://docs.google.com/document/d/1j41XUAGe6GxzPMfSX0K3d4n-aUc3QI23hbwz_TaWBN4/edit?usp=sharing

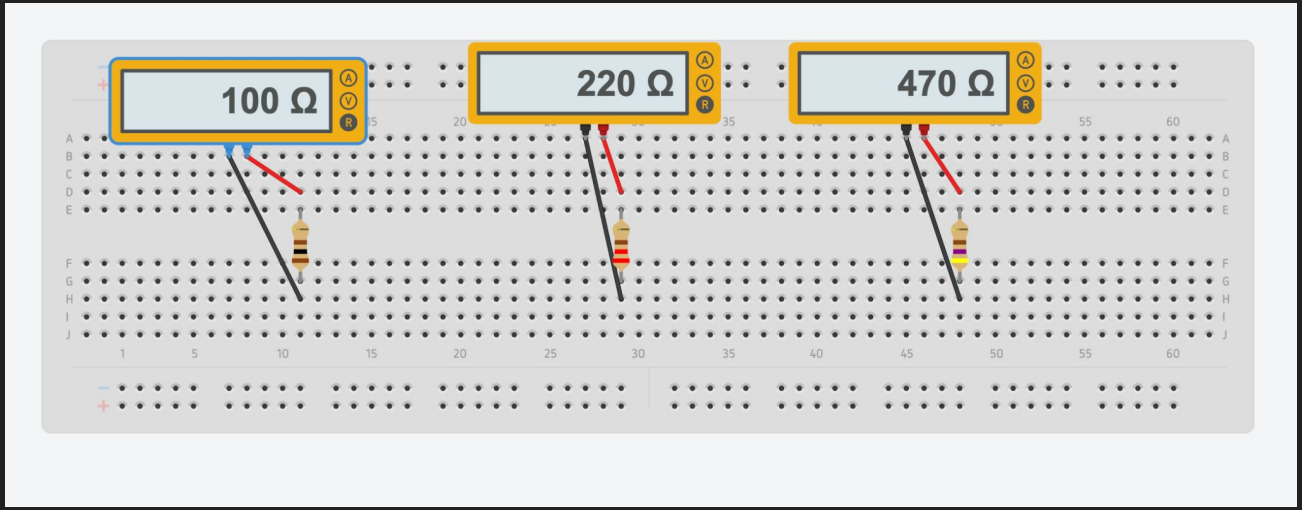
2. Build a circuit to replicate the values in questions 2 and 3 of your worksheet. Look at the power supply and determine if the results match your calculations for current. Embed your circuit screenshot(s) on the next 2 slides. Also include the URLs to your tinkercad projects.


22.8 V
5.00 A
ON
5.00 A
A
V
R


1.50 V
10.0 mA
ON
10.0 mA
A
V
Ω
1 5 10 15 20 25 30 35 40 45 50 55 60
A B C D E
F G H I J

Questions and Exercises

3. Build a circuit with no power. Add 3 different resistors of varying values. Use three multimeters to measure their resistance. Embed your live circuit screenshot below.



Questions and Exercises

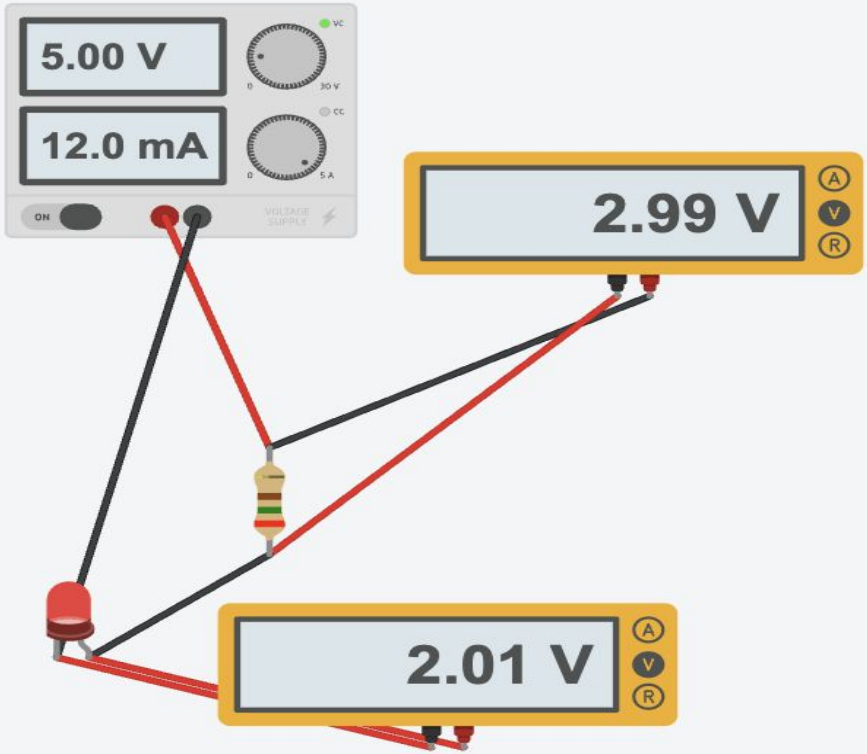
4. Create a new circuit with a single resistor at 500 Ohms. Connect the resistor to a power supply at 5V. Measure the voltage across the resistor. Add in an LED to the circuit. Now measure the voltage across the resistor. What happens? Why?The voltage across the resistor decreases.The circuit is in series, and the voltage is distributed according to the resistance of the components.

Measure the voltage across the LED. What is its value? What is the sum of the voltages across the resistor and the LED? 1.94V, 5V.

Decrease the resistance to 250 Ohms. Measure the voltages across the LED and the resistor. What happens to their values? What is the sum of the voltages?When the resistor voltage decreases, the LED voltage increases.

What can you conclude from this? Include a live circuit screenshot in the next slide.

In a series circuit, the total voltage remains constant.

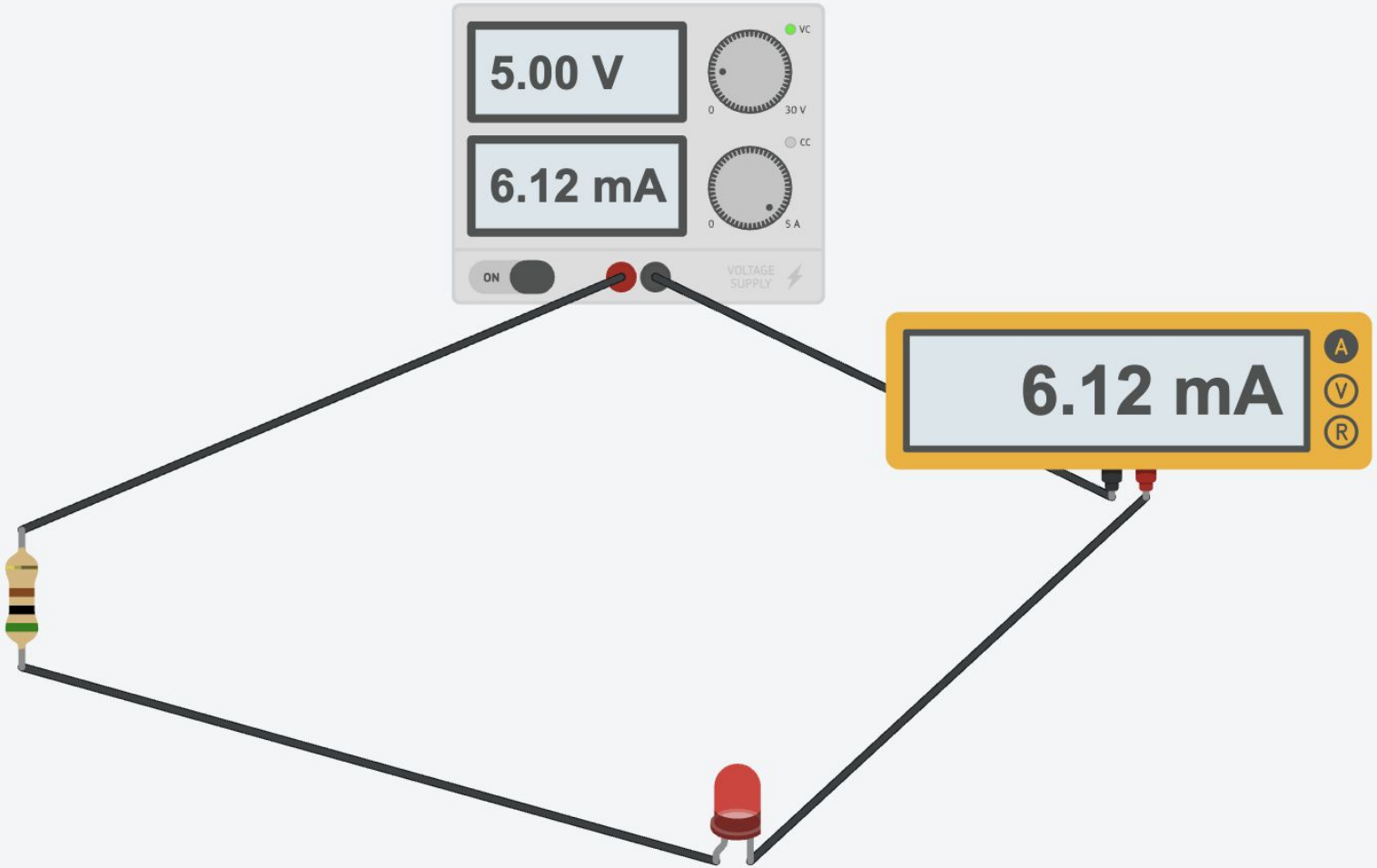

5.00 V
12.0 mA
ON
2.99 V
A
V
R
2.01 V
A
V
R

Questions and Exercises

5. Create a new circuit with a single resistor at 500 Ohms. Connect the resistor to a power supply at 5V. Measure the current in the circuit using the multimeter. What is its value? 0.1A Add in an LED. Measure the current at 3 points in the circuit and list their values: 1. Just before the resistor 6.12mA 2. Just after the resistor and before the LED 6.12mAand 3. Just after the LED. 6.12mA What do you notice about their values?no change

Decrease the resistors value to 250 Ohms and repeat the measurements. What do you notice?no change

Include a live circuit screenshot in the next slide.


5.00 V
6.12 mA
VC
0
30 V
CC
0
5 A
ON
VOLTAGE SUPPLY
6.12 mA
A
V
R